

# Probability Theory

*Lecture Notes*

---

Instructor: Dr. Nishant Chandgotia

---

Written By-

**Kartick Samanta**

 **Centre For Applicable Mathematics**

Fall 2026

---

2026-09-07

# Probability Theory

Lecture Notes

---

Fall 2026

---

**Student** Kartick Samanta

**Course** Probability Theory

**Instructor** Dr. Nishant Chandgotia

**University** TIFR- CAM

2026-09-07

# Contents

|                                                                      |           |
|----------------------------------------------------------------------|-----------|
| <b>1.1 Motivation and Some Problems</b>                              | <b>1</b>  |
| <b>2.1 Some Introductory Idea</b>                                    | <b>4</b>  |
| <b>2.2 Quiz, 10 August 2026</b>                                      | <b>7</b>  |
| <b>3.1 Central Limit Theorem and Convergence</b>                     | <b>8</b>  |
| 3.1.1 Galton Board . . . . .                                         | 8         |
| 3.1.2 Random Walk . . . . .                                          | 9         |
| 3.1.3 Convergence in Distribution . . . . .                          | 9         |
| 3.1.4 An Example . . . . .                                           | 9         |
| <b>3.2 Objects in Probability</b>                                    | <b>10</b> |
| <b>3.3 Sample Space and Sigma Algebra</b>                            | <b>11</b> |
| 3.3.1 Sigma Algebra . . . . .                                        | 12        |
| 3.3.2 Cylinder Sets . . . . .                                        | 12        |
| <b>4.1 Sample Spaces, Random Variables and Their Distributions</b>   | <b>13</b> |
| 4.1.1 Sample Space . . . . .                                         | 13        |
| 4.1.2 $\sigma$ -Algebra . . . . .                                    | 13        |
| 4.1.3 Probability Measure . . . . .                                  | 13        |
| 4.1.4 Non-measurable Sets . . . . .                                  | 14        |
| <b>4.2 Random Variables</b>                                          | <b>15</b> |
| 4.2.1 A Random Variable Detecting the Existence of a Limit . . . . . | 15        |
| <b>4.3 Distribution of a Random Variable</b>                         | <b>16</b> |
| <b>4.4 Some Useful Definitions:</b>                                  | <b>16</b> |
| 4.4.1 Convergence in Probability . . . . .                           | 16        |
| 4.4.2 Convergence in Distribution . . . . .                          | 17        |
| 4.4.3 Convergence in $L^1$ . . . . .                                 | 17        |
| 4.4.4 Almost Sure Convergence . . . . .                              | 17        |
| <b>4.5 Types of Random Variables</b>                                 | <b>18</b> |

|                                                                    |           |
|--------------------------------------------------------------------|-----------|
| 4.5.1 Examples of Discrete Random Variables . . . . .              | 18        |
| <b>4.6 Quiz, 17 August 2026</b>                                    | <b>19</b> |
| <b>5.1 Continuous Random Variables</b>                             | <b>20</b> |
| 5.1.1 Continuous Random Variables . . . . .                        | 20        |
| 5.1.2 Absolutely Continuous and Singular Distributions . . . . .   | 20        |
| 5.1.3 Examples of Continuous Random Variables . . . . .            | 21        |
| 5.1.4 A Singular Continuous Distribution . . . . .                 | 22        |
| 5.1.5 The Cantor Distribution . . . . .                            | 22        |
| 5.1.6 Bernoulli Convolution . . . . .                              | 23        |
| <b>5.2 Bertrand's Paradox</b>                                      | <b>24</b> |
| 5.2.1 Method 1: Choosing Two Endpoints . . . . .                   | 24        |
| 5.2.2 Method 2: Choosing the Midpoint . . . . .                    | 24        |
| <b>5.3 Distribution as a Push-Forward</b>                          | <b>25</b> |
| 5.3.1 Change of Variables . . . . .                                | 25        |
| 5.3.2 Example: Chord Length . . . . .                              | 26        |
| <b>6.1 Distribution Functions and Random Variables</b>             | <b>28</b> |
| 6.1.1 Properties of the C.D.F. . . . .                             | 29        |
| <b>6.2 The Probability Measure Associated with a C.D.F.</b>        | <b>29</b> |
| 6.2.1 Finite Additivity . . . . .                                  | 30        |
| 6.2.2 Countable Additivity . . . . .                               | 30        |
| <b>6.3 Correspondence Between C.D.F.s and Probability Measures</b> | <b>31</b> |
| <b>6.4 Constructing a Random Variable from a C.D.F.</b>            | <b>31</b> |
| 6.4.1 Case 1: $F$ is Continuous and Strictly Increasing . . . . .  | 32        |
| 6.4.2 Case 2: $F$ Has a Jump . . . . .                             | 32        |
| 6.4.3 Case 3: $F$ Has a Flat Portion . . . . .                     | 33        |
| 6.4.4 Case 4: General C.D.F. . . . .                               | 33        |
| <b>6.5 Quiz, 24 August 2026</b>                                    | <b>35</b> |
| <b>7.1 Examples of Random Variables</b>                            | <b>36</b> |

|                                              |    |
|----------------------------------------------|----|
| 7.2A Gaussian Tail Estimate                  | 37 |
| 7.3 Expectation and Variance                 | 37 |
| 7.4 Central Limit Theorem                    | 38 |
| 7.5 Examples of Expectation and Variance     | 38 |
| 7.6A Result on Signs of Vectors              | 39 |
| 7.7 Exercises on Probability Distributions   | 40 |
| 7.8 Graph Theory                             | 40 |
| 7.9 Bipartite Graphs                         | 41 |
| 7.10 Subgraphs and Walks                     | 41 |
| 7.11 Trees and Bipartite Graphs              | 41 |
| 7.12 Characterization of Bipartite Graphs    | 42 |
| 7.13 Bipartite Subgraph                      | 43 |
| 8.1 Inequalities                             | 46 |
| 8.2 Coupon Collector Problem                 | 47 |
| 8.3 Quiz, 4 September 2026                   | 49 |
| 9.1 Continuation of Coupon Collector Problem | 50 |
| 9.2 Erdős–Rényi Graphs                       | 52 |
| 9.2.1 Full Strength of Chebyshev . . . . .   | 54 |

# Probability Theory

## Lecture 01

### Introduction to Probability

**Date:** Wednesday, 05 August 2026

**Instructor:** Dr. Nishant Chandgotia

## 1.1 Motivation and Some Problems

### Question 1. Sleeping Beauty Problem

A princess is put into a deep sleep by a powerful drug administered by a villain. The drug ensures that she will remain asleep for two years, except for possible brief awakenings determined by the outcome of a fair coin toss.

The villain tosses a fair coin.

- If the coin lands **Heads**, the princess is awakened *once*, briefly, at the end of the **first year**. After this awakening, her memory of having awakened is completely erased, and she is put back to sleep until the end of the second year.
- If the coin lands **Tails**, the princess is not awakened after the first year. Instead, she is awakened *once*, briefly, at the end of the **second year**.

Whenever the princess awakens, she has no information about the current date or the passage of time. Furthermore, if she had awakened previously, all memory of that awakening has been erased. Consequently, upon waking, she cannot distinguish between the two possible awakening scenarios.

Suppose that the princess wakes up and is immediately asked the following question:

*“What probability should you assign to the event that the coin landed Heads?”*

What should the princess answer?

Solution:

1.  $\frac{1}{2}$  No information exchanged so probability should be  $\frac{1}{2}$ , If repeated independently a million times, half chances each.
2.  $\frac{2}{3}$  There are possibilities of this. Proportion of waking during Heads is  $\frac{2}{3}$ , and  $\dots \frac{1}{3}$
3.  $\frac{3}{4}$

Villain tosses once, H  $\rightarrow$  Villain tosses again, Two Possibilities H and T, so,

- HH  $\rightarrow \frac{1}{4}$
- HT  $\rightarrow \frac{1}{4}$

- $T \rightarrow \frac{1}{2}$

### Important

All the solutions above are wrong. The Above problem has no solution, Because it is ill posed, and this ill posedness is with respect to current Framework.

### Question 2. Monty Hall Problem

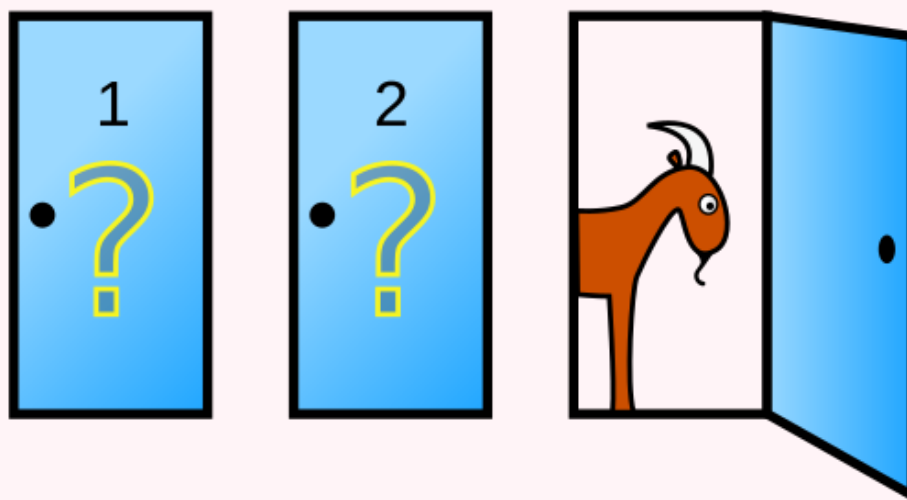


Figure 1: Monty Hall Problem

There are three doors: one hides a car and the other two hide goats. After you choose one door say A, the host opens another door revealing a goat.

1. Question: Should you switch?
2. Probability car behind C > Probability car behind A.

Solution: Yes, we should change our decision,

Initially, the probability that your chosen door contains the car is  $P(\text{chosen door}) = \frac{1}{3}$ , while the probability that the car is behind one of the other two doors is  $P(\text{other two doors}) = \frac{2}{3}$ .

Since the host always reveals a goat, this entire probability concentrates on the remaining unopened door. Therefore,  $P(\text{Win by staying}) = \frac{1}{3}$ ,  $P(\text{Win by switching}) = \frac{2}{3}$ . Hence, switching gives twice the probability of winning compared to staying.

### Remark

The Above question (Monty Hall Problem) is not ill posed.

## Summary

- Sleeping Beauty Problem
- Monty Hall Problem

---

End of Lecture

---

# Probability Theory

## Lecture 02

### Introduction to Probability

**Date:** Friday, 07 August 2026

**Instructor:** Dr. Nishant Chandgotia

## 2.1 Some Introductory Idea

Suppose we toss a fair coin  $n$ -times.  $HTH HTTH \dots T$

What is the Probability of the above sequence ? Its-  $\sim \frac{1}{2^n}$

If we have all sequences with  $k$ -heads then  $P(k\text{-heads}) = \frac{\binom{n}{k}}{2^n}$ , and it will be maximum when  $k = \frac{n}{2}$ .

If  $x_1, x_2, \dots, x_k$  are randomly independently chosen, then  $\sum_{i=1}^k x_i \sim \frac{r+1}{2} \cdot k$ , if they are chosen from the uniform distribution on  $\{1, 2, \dots, r\}$ .

Now Stirling's Formula,  $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$ , for very large  $n$ .

Now,

$$\frac{P(\text{number of heads} = k)}{P(\text{number of heads} = k+1)} = \frac{k+1}{n-k}.$$

So,  $\frac{k+1}{n-k} > 1 \iff k > \frac{n-1}{2}$ , and  $\frac{k+1}{n-k} < 1 \iff k < \frac{n-1}{2}$ .

As  $\binom{n}{k}$  is maximum when  $k = \frac{n}{2}$ , therefore,  $\frac{\binom{n}{n/2}}{2^n} \sim \frac{2}{\sqrt{2\pi n}}$  (By using Stirling's Formula). It

would be unjustified to say  $\# \text{Heads} = \frac{n}{2}$  with high probability.

Suppose  $k = \alpha n$ ,  $\alpha \neq \frac{1}{2}$ .

Then  $\frac{\binom{n}{k}}{2^n} \sim \frac{1}{\sqrt{2\pi n}} e^{n(-\alpha \log \alpha - (1-\alpha) \log(1-\alpha) - \log 2)}$ .

Let,  $H(\alpha) = -\alpha \log \alpha - (1-\alpha) \log(1-\alpha)$ , be Boltzman Shanon Entropy (Mathematical theory of Communications).

Hence,

$$\frac{\binom{n}{k}}{2^n} \sim \frac{1}{\sqrt{2\pi n}} e^{n(H(\alpha) - H(\frac{1}{2}))}.$$

Now,  $H(\alpha) \leq H(\frac{1}{2})$ , by using Jensen's inequality as  $\log$  is a concave function.

Equivalently,  $-\alpha \log \alpha - (1-\alpha) \log(1-\alpha) < \ln 2$ .

Therefore,  $\frac{\binom{n}{k}}{2^n} \rightarrow 0$  exponentially.

Now,

$$P(\#\text{Heads} \in ((\alpha - \varepsilon)n, (\alpha + \varepsilon)n)) = \sum_{k=(\alpha-\varepsilon)n}^{(\alpha+\varepsilon)n} \frac{\binom{n}{k}}{2^n}, \quad \text{where } \varepsilon > 0, \quad \frac{1}{2} \notin (\alpha - \varepsilon, \alpha + \varepsilon).$$

Also,

$$\frac{2n\varepsilon}{2^n} \binom{n}{(\alpha - \varepsilon)n} \leq \cdots \leq \frac{2n\varepsilon}{2^n} \binom{n}{(\alpha + \varepsilon)n}.$$

Therefore, the asymptotic bounds are  $\frac{2n\varepsilon}{\sqrt{2\pi n}} e^{n(H(\alpha-\varepsilon)-H(\frac{1}{2}))}$  and  $\frac{2n\varepsilon}{\sqrt{2\pi n}} e^{n(H(\alpha+\varepsilon)-H(\frac{1}{2}))}$ .

Therefore,  $P(\#\text{Heads} \in ((\alpha - \varepsilon)n, (\alpha + \varepsilon)n)) \rightarrow 0$  exponentially as  $n \rightarrow \infty$ .

Hence,

$$P\left(\#\text{Heads} \in n\left(\frac{1}{2} - \varepsilon, \frac{1}{2} + \varepsilon\right)\right) \rightarrow 1.$$

This is called the **Large Deviation Phenomenon**.

$P(\text{staying close to mean}) \rightarrow 1$ ,

whereas  $P(\text{deviating from the mean significantly}) \rightarrow 0$  exponentially.

In our case, the rate function is  $H(\alpha) - H(\frac{1}{2})$ .

Let  $f_n = \frac{\#\text{Heads}}{n}$ . Then  $f_n \rightarrow f$  in measure.

Therefore,

$$P\left(\left|\frac{\#\text{Heads}}{n} - \frac{1}{2}\right| > \varepsilon\right) \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (1)$$

We call this **convergence in probability**.

Later we shall use the **Borel–Cantelli Lemma** to prove from (1) that  $\frac{\#\text{Heads}}{n} \rightarrow \frac{1}{2}$  almost surely.

The sample space is  $\Omega = \{H, T\}^{\mathbb{N}}$ . We equip  $\Omega$  with the product topology, where each copy of  $\{H, T\}$  carries the discrete topology. The corresponding Borel  $\sigma$ -algebra,  $\mathcal{F} = \mathcal{B}(\Omega)$ , is generated by the cylinder sets. The basis of the product topology consists of the cylinder sets, namely sets of the form  $\{\omega \in \Omega : \omega_{i_1} = a_1, \dots, \omega_{i_k} = a_k\}$ , where  $i_1, \dots, i_k$  are finitely many coordinates and each  $a_j \in \{H, T\}$ .

If we only specify the first  $n$  coordinates, they look like  $\{\omega : \omega_1 = a_1, \dots, \omega_n = a_n\}$ , which are the basic cylinders.

The complete probability space is  $(\Omega, \mathcal{F}, \mu) = (\{H, T\}^{\mathbb{N}}, \mathcal{B}(\{H, T\}^{\mathbb{N}}), \mu)$ ,

Recall,

$$P\left(\#\text{Heads} = \frac{n}{2}\right) \sim \frac{2}{\sqrt{2\pi n}}.$$

Suppose

$$c > 0,$$

and let  $|k - \frac{n}{2}| < c\sqrt{n}$ .

Then

$$P(\text{\#Heads} = k) = \frac{n!}{(n-k)!k!} \cdot \frac{1}{2^n}.$$

Using Stirling's formula,

$$\begin{aligned} P(\text{\#Heads} = k) &\sim \frac{1}{\sqrt{\pi n/2}} \cdot \frac{1}{\left(2 - \frac{2k}{n}\right)^{n-k} \left(\frac{2k}{n}\right)^k} \\ &= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \cdot \frac{1}{\left(1 + \left(1 - \frac{2k}{n}\right)\right)^{n-k} \left(1 + \left(\frac{2k}{n} - 1\right)\right)^k}. \end{aligned}$$

Hence,

$$= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \exp\left(-\frac{(n-2k)^2}{n}\right).$$

Equivalently,  $= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \exp\left(-\frac{(k - \frac{n}{2})^2}{2 \cdot \frac{n}{4}}\right)$ , which resembles the normal distribution.

$$\mu = \frac{1}{2}, \quad \sigma = \frac{1}{2}.$$

Hence,

$$= \frac{1}{\sqrt{2\pi\sigma^2n}} \exp\left(-\frac{(k - \mu n)^2}{2\sigma^2n}\right).$$

We can prove,  $P\left(a < \frac{\text{\#Heads} - \mu n}{\sigma\sqrt{n}} < b\right) \longrightarrow \int_a^b \varphi(x) dx$ , as  $n \rightarrow \infty$ . (**Central Limit Theorem / De Moivre–Laplace**)

---

End of Lecture

---

## 2.2 Quiz, 10 August 2026

### Quiz 1

1. Let  $p < \frac{1}{2}$ . Suppose we toss  $n$  coins independently such that the probability of heads is  $p$  and tails is  $q = 1 - p$ .

For what value of  $k$  is the probability of getting  $k$  heads maximised?

2. For probability vectors  $P = (p_1, p_2, \dots, p_n)$ ,  $Q = (q_1, q_2, \dots, q_n)$ , where  $p_i, q_i > 0$ ,  $\sum_i p_i = \sum_i q_i = 1$ , define the Kullback–Leibler divergence by

$$D_{KL}(P\|Q) = \sum_i p_i \log \left( \frac{p_i}{q_i} \right).$$

Use Jensen's inequality to show that  $D_{KL}(P\|Q) \geq 0$  with equality if and only if  $P = Q$ .

3. Let  $\alpha \in (0, 1)$ . Prove that the number of heads equalling  $n\alpha$  is asymptotically

$$\frac{1}{\sqrt{2\pi n\alpha(1-\alpha)}} \exp(-nD_{KL}(P\|Q)),$$

where  $P = (\alpha, 1 - \alpha)$ ,  $Q = (p, q)$ .

# Probability Theory

## Lecture 03

### Introduction to Probability

**Date:** Wednesday, 12 August 2026

**Instructor:** Dr. Nishant Chandgotia

---

## 3.1 Central Limit Theorem and Convergence

Suppose we toss a fair coin independently  $n$  times. The Central Limit Theorem suggests that

$$P\left(a < \frac{\#\text{Heads} - \mu n}{\sigma\sqrt{n}} < b\right) \longrightarrow \int_a^b \varphi(x) dx, \quad n \rightarrow \infty,$$

where, for a fair coin, and

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

### 3.1.1 Galton Board

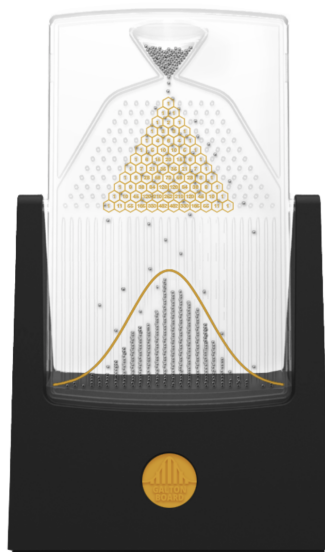


Figure 2: Galton board. Source: [Wikimedia Commons](#).

The Galton board gives a visual representation of a process whose distribution approaches a normal distribution.

### 3.1.2 Random Walk

A *random walk* is a stochastic process in which the position changes randomly at each step.

For example, consider a person starting at the origin. At every step, the person tosses a fair coin:

$$\begin{cases} H & \longrightarrow \text{move one step to the right,} \\ T & \longrightarrow \text{move one step to the left.} \end{cases}$$

This gives the classical one-dimensional symmetric random walk.

There are several natural questions:

#### Questions

- Will the person return to the origin?
- Will the person escape to infinity?
- How often will the person return to the origin?

### 3.1.3 Convergence in Distribution

We know,

$$\frac{\#\text{Heads}}{n} \longrightarrow \frac{1}{2}.$$

Now define

$$f_n = \frac{\#\text{Heads} - \mu n}{\sigma \sqrt{n}}.$$

Then the Central Limit Theorem states that

$$P(a < f_n < b) \longrightarrow \int_a^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx.$$

This leads to the notion of *weak convergence*, or *convergence in distribution*.

A natural question is whether convergence in distribution implies almost-sure convergence.

#### Important

Convergence in distribution does not, in general, imply almost-sure convergence.

### 3.1.4 An Example

Consider the following random process.

Start with 1.

Toss a fair coin independently at every step:

$$\begin{cases} H & \longrightarrow \text{multiply by } 2, \\ T & \longrightarrow \text{multiply by } -2. \end{cases}$$

After  $n$  steps, define

$$f_n = \frac{\text{product after } n \text{ steps}}{2^n}.$$

Since every multiplication contributes either 2 or  $-2$ ,

$$f_n \in \{-1, 1\}.$$

Moreover,

$$P(f_n = 1) = \frac{1}{2}, \quad P(f_n = -1) = \frac{1}{2}.$$

Thus the distribution of  $f_n$  is the same for every  $n$ .

However,  $f_n$  does not converge almost surely.

## Questions

- Does  $f_n$  converge almost surely?
- Does  $f_n$  converge in  $L^1$ ?
- Does  $f_n$  converge in probability?

### Remark

The example illustrates that convergence in distribution is a weaker notion of convergence than almost-sure convergence.

## 3.2 Objects in Probability

There is a close analogy between measure theory and probability theory.

Table 1: Analogy between Measure Theory and Probability

| Measure Theory                                                                     | Probability                                                        |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Measure space: $(\Omega, \mathcal{B}, \mu)$                                        | Probability space: $(\Omega, \mathcal{B}, \mathbb{P})$             |
| $\sigma$ -algebra $\mathcal{B}$ : $A \in \mathcal{B}$ , where $\mu(A)$ is defined. | Event: $A \in \mathcal{B}$ , for which $\mathbb{P}(A)$ is defined. |
| Elements: $\omega \in \Omega$                                                      | Outcomes / samples: $\omega \in \Omega$                            |
| Measurable function:                                                               | Random variable:                                                   |
| $f : (\Omega, \mathcal{B}) \longrightarrow (S, \rho)$                              | $X : (\Omega, \mathcal{B}, \mathbb{P}) \longrightarrow (S, \rho)$  |
| $L^1$ -integrability:                                                              | Mean / expectation:                                                |
| $\int_{\Omega}  f  d\mu < \infty$                                                  | $\mathbb{E}[X] = \int_{\Omega} X d\mathbb{P}$                      |
| $L^2$ -norm:                                                                       | Variance:                                                          |
| $\ f\ _{L^2}^2 = \int_{\Omega}  f ^2 d\mu$                                         | $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$                |
| Push-forward measure:                                                              | Distribution of $X$ :                                              |
| $f_*\mu(A) = \mu(f^{-1}(A))$                                                       | $\nu(A) = \mathbb{P}(X^{-1}(A))$                                   |
| Fourier transform:                                                                 | Characteristic function:                                           |
| $\hat{f}(t) = \int_{\mathbb{R}} e^{itx} f(x) dx$                                   | $\Phi_X(t) = \mathbb{E}[e^{itX}]$                                  |
| Convergence in measure                                                             | Convergence in probability                                         |
| Almost everywhere (a.e.)                                                           | Almost surely (a.s.)                                               |
| Convergence in $L^1$                                                               | Convergence in mean.                                               |

### 3.3 Sample Space and Sigma Algebra

The first object in probability is the *sample space*.

$\Omega$  = the universe of all possible outcomes.

For example, for a single coin toss,  $\Omega = \{H, T\}$ .

For a bus arriving at a non-negative time,  $\Omega = [0, \infty)$ .

For infinitely many coin tosses,  $\Omega = \{H, T\}^{\mathbb{N}}$ .

An element of this space is an infinite sequence  $\omega = (\omega_1, \omega_2, \dots)$ ,  $\omega_i \in \{H, T\}$ .

### 3.3.1 Sigma Algebra

Let  $\mathcal{B}$  denote the  $\sigma$ -algebra of events.

For a single coin toss,  $\mathcal{B} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$ .

For the infinite product space  $\{H, T\}^{\mathbb{N}}$ , the relevant  $\sigma$ -algebra is generated by cylinder sets.

### 3.3.2 Cylinder Sets

Suppose we are interested only in the first finitely many tosses.

Let  $F \subset \mathbb{N}$  be a finite set and let  $a : F \rightarrow \{H, T\}$  be a fixed assignment of heads and tails.

Define  $[a] = \{\omega \in \{H, T\}^{\mathbb{N}} : \omega_i = a(i) \text{ for every } i \in F\}$ .

The set  $[a]$  specifies the outcomes of the coordinates belonging to the finite set  $F$ , while leaving all other coordinates unrestricted.

Such sets are called *cylinder sets*.

They form a basis for the product topology on  $\{H, T\}^{\mathbb{N}}$ .

We want to be able to assign a probability  $\mathbb{P}([a])$  to every cylinder set.

For a fair coin, if  $|F| = k$ , then  $\mathbb{P}([a]) = \frac{1}{2^k}$ .

---

End of Lecture

---

# Probability Theory

## Lecture 04

### Introduction to Probability

**Date:** Friday, 14 August 2026

**Instructor:** Dr. Nishant Chandgotia

## 4.1 Sample Spaces, Random Variables and Their Distributions

### 4.1.1 Sample Space

The first object in probability is the *sample space*  $\Omega$ , which is the collection of all possible outcomes.

1. **Simple coin toss**  $\Omega = \{H, T\}$ .
2. **Arrival time of a bus** Depending on the context, we may take  $\Omega = [0, 1]$  or  $\Omega = [0, \infty)$ .
3. **Infinite sequence of coin tosses**  $\Omega = \{H, T\}^{\mathbb{N}}$ .

### 4.1.2 $\sigma$ -Algebra

1. A  $\sigma$ -algebra  $\mathcal{B}$  is the collection of events whose probabilities we can measure.  
For a simple coin toss,  $\mathcal{B} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$ .
2. For a subset of  $\mathbb{R}$ , we are naturally led to the *Borel  $\sigma$ -algebra*.
3. For the infinite product space  $\{H, T\}^{\mathbb{N}}$ , consider a finite set  $F \subset \mathbb{N}$  and a function  $a : F \rightarrow \{H, T\}$ .

Define the corresponding cylinder set by

$$[a] = \left\{ \omega \in \{H, T\}^{\mathbb{N}} : \omega_i = a(i) \text{ for every } i \in F \right\}.$$

Cylinder sets form the basic building blocks for the product  $\sigma$ -algebra.

### 4.1.3 Probability Measure

A probability measure is a measure  $\mathbb{P} : \mathcal{B} \rightarrow [0, 1]$  such that  $\mathbb{P}(\Omega) = 1$ .

1. For a fair coin,  $\mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) = \frac{1}{2}$ .

2. For the uniform probability measure on  $[0, 1]$ ,

$$\mathbb{P}([a, b]) = b - a, \quad 0 \leq a \leq b \leq 1.$$

3. For independent coin tosses, if  $[a]$  is a cylinder set determined by  $|F|$  coordinates, then

$$\mathbb{P}([a]) = 2^{-|F|}.$$

4. More generally, for a coin with  $\mathbb{P}(H) = p$ ,  $\mathbb{P}(T) = 1 - p$ , we have

$$\mathbb{P}([a]) = p^{\#\text{Heads}}(1 - p)^{\#\text{Tails}}.$$

Question: Why does this define a probability measure on  $\{H, T\}^{\mathbb{N}}$ ?

## 4.1.4 Non-measurable Sets

### Important

Not every subset of  $\{H, T\}^{\mathbb{N}}$  is measurable.

The construction of non-measurable sets is related to the existence of suitable equivalence relations and translation-invariant measures.

A useful analogy is the classical construction for  $\mathbb{R}$ .

### 4.1.4.1 Proof Idea on $\mathbb{R}$

The general strategy is:

1. Find a countable subgroup of  $\mathbb{R}$ .
2. Find a Borel measure satisfying translation invariance:

$$\mu(A + x) = \mu(A), \quad \forall x \in \mathbb{R}, \quad A \in \mathcal{B}.$$

### 4.1.4.2 Analogue for the Infinite Coin-Toss Space

Consider  $\left(\frac{\mathbb{Z}}{2\mathbb{Z}}\right)^{\mathbb{N}}$ .

One can consider the set

$$G = \left\{ x \in \left(\frac{\mathbb{Z}}{2\mathbb{Z}}\right)^{\mathbb{N}} : x(i) = 0 \text{ for all sufficiently large } i \right\}.$$

The idea is to check that  $G$  is a countable subgroup and then investigate translation invariance of the probability measure on cylinder sets.

In particular, for a cylinder set  $[a]$ ,

$$\mathbb{P}([a]) = 2^{-|F|}.$$

## 4.2 Random Variables

A random variable is a measurable function from the probability space to another measurable space.

For example, for infinitely many coin tosses,  $f_n : \{H, T\}^{\mathbb{N}} \rightarrow [0, 1]$ , defined by

$$f_n(\omega) = \frac{1}{n} \sum_{i=1}^n X_i(\omega),$$

where

$$X_i(\omega) = \begin{cases} 1, & \omega_i = H, \\ 0, & \omega_i = T. \end{cases}$$

We would like to understand the limit

$$f(\omega) = \lim_{n \rightarrow \infty} f_n(\omega).$$

### 4.2.1 A Random Variable Detecting the Existence of a Limit

#### Question 1. Random Variable for Existence of Limit

Given a probability space and a sequence of random variables  $(f_n)$ , define a random variable which captures whether or not  $\lim_{n \rightarrow \infty} f_n(\omega)$  exists.

For a real-valued sequence, the limit exists precisely when  $\limsup_{n \rightarrow \infty} f_n(\omega) = \liminf_{n \rightarrow \infty} f_n(\omega)$  and both quantities are finite.

Define

$$\tilde{f}(\omega) = \begin{cases} \limsup_{n \rightarrow \infty} f_n(\omega) - \liminf_{n \rightarrow \infty} f_n(\omega), & \text{if both are finite,} \\ +\infty, & \text{otherwise.} \end{cases}$$

Then define

$$f'(\omega) = \begin{cases} 1, & \tilde{f}(\omega) = 0, \\ 0, & \tilde{f}(\omega) \neq 0. \end{cases}$$

Thus

$$f'(\omega) = \mathbf{1}_{\{\lim_{n \rightarrow \infty} f_n(\omega) \text{ exists as a finite real number}\}}.$$

#### Remark

The precise formulation and notation may differ from the presentation in Durrett.

## 4.3 Distribution of a Random Variable

Let  $X : (\Omega, \mathcal{B}, \mathbb{P}) \longrightarrow (S, \rho)$  be a random variable.

The distribution of  $X$  is the push-forward of  $\mathbb{P}$  under  $X$ :  $X_*\mathbb{P}$ .

More explicitly, the distribution  $\mu_X$  of  $X$  is the measure on  $(S, \rho)$  defined by

$$\mu_X(A) = \mathbb{P}\left(X^{-1}(A)\right), \quad A \in \rho.$$

It is also common to write

$$\mathbb{P}(X \in A) := \mathbb{P}\left(X^{-1}(A)\right).$$

Consider the random variables  $X_1$  and  $X_2$  arising from the independent coin-toss example above.

Then  $X_1$  and  $X_2$  have the same distribution:  $X_1 \stackrel{d}{=} X_2$ .

However, they are not the same random variable.

Indeed,  $X_1(\omega) \neq X_2(\omega)$  for some  $\omega$ .

Also,  $(X_1, X_2)$  is itself a random variable, taking values in the appropriate product space.

### Remark

Having the same distribution does not imply that two random variables are equal.  
In symbols,

$$X = Y \implies X \stackrel{d}{=} Y,$$

but in general,

$$X \stackrel{d}{=} Y \not\implies X = Y.$$

## 4.4 Some Useful Definitions:

Let  $(X_n)_{n \geq 1}$  and  $X$  be random variables defined on a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ .

### 4.4.1 Convergence in Probability

#### Definition 1. Convergence in Probability

We say that  $X_n$  **converges in probability** to  $X$ , denoted by

$$X_n \xrightarrow{\mathbb{P}} X,$$

if for every  $\varepsilon > 0$ ,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0.$$

Equivalently, for every  $\varepsilon > 0$ ,

$$\mathbb{P}(|X_n - X| > \varepsilon) \longrightarrow 0.$$

#### 4.4.2 Convergence in Distribution

##### Definition 2. Convergence in Distribution

We say that  $X_n$  **converges in distribution** (or *in law*) to  $X$ , denoted by

$$X_n \xrightarrow{d} X,$$

if

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

for every continuity point  $x$  of the distribution function  $F_X$  of  $X$ .

Here,

$$F_{X_n}(x) = \mathbb{P}(X_n \leq x), \quad F_X(x) = \mathbb{P}(X \leq x).$$

#### 4.4.3 Convergence in $L^1$

##### Definition 3. $L_1$ Convergence

We say that  $X_n$  **converges to  $X$  in  $L^1$** , denoted by

$$X_n \xrightarrow{L^1} X,$$

if

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n - X|] = 0.$$

Equivalently,

$$\|X_n - X\|_1 = \mathbb{E}[|X_n - X|] \longrightarrow 0.$$

#### 4.4.4 Almost Sure Convergence

##### Definition 4. Almost Sure Convergence

We say that  $X_n$  **converges almost surely** to  $X$ , denoted by

$$X_n \xrightarrow{\text{a.s.}} X,$$

if

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\right\}\right) = 1.$$

Equivalently,

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1.$$

That is, the sequence  $X_n(\omega)$  converges to  $X(\omega)$  for almost every  $\omega \in \Omega$ .

## 4.5 Types of Random Variables

Random variables can be classified according to their distributions.

1. **Discrete random variables**
  2. **Continuous random variables**
    - (a) Absolutely continuous random variables.
    - (b) Singular random variables.
- These classifications are determined by the distribution of the random variable.
  - A discrete random variable has an *atomic distribution*.

### 4.5.1 Examples of Discrete Random Variables

1. **Bernoulli random variable** Let  $p \in [0, 1]$ . We write  $X \sim \text{Ber}(p)$  if

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p.$$

This notation means that  $X$  has the Bernoulli distribution with parameter  $p$ .

2. **Binomial random variable**

Let  $n \in \mathbb{N}$ ,  $p \in [0, 1]$ . We write  $X \sim \text{Bin}(n, p)$

if

$$\mathbb{P}(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}, \quad r = 0, 1, \dots, n.$$

3. **Poisson random variable**

Let  $\lambda > 0$ . We write  $X \sim \text{Poi}(\lambda)$

if

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

4. **Geometric random variable**

Let  $p \in (0, 1)$ . Under the convention that  $X$  takes values in  $\{1, 2, \dots\}$ , we write  $X \sim \text{Geo}(p)$

if

$$\mathbb{P}(X = k) = p(1 - p)^{k-1}, \quad k = 1, 2, \dots$$

---

End of Lecture

---

## 4.6 Quiz, 17 August 2026

### Quiz 2

1. Convergence in probability implies convergence almost surely.
2. Convergence in distribution implies convergence in probability.
3. Convergence almost surely implies convergence in distribution.
4. Let  $X, Y, Z \sim \text{Ber}(1/2)$  such that  $X \perp Y$ ,  $Y \perp Z$  and  $Z \perp X$ . Then

$$\mathbb{P}(X = 1, Y = 1, Z = 1) = \frac{1}{8}.$$

# Probability Theory

## Lecture 05

### Introduction to Probability

**Date:** Wednesday, 19 August 2026

**Instructor:** Dr. Nishant Chandgotia

## 5.1 Continuous Random Variables

### Question 1. Quiz-Question-Discussion

The last question from the quiz showed that pairwise independence and mutual independence are different notions. What is the distinction between the two?

### 5.1.1 Continuous Random Variables

Let  $X : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (S, \rho)$  be a random variable.

In many cases, the state space is one of  $\mathbb{R}$ ,  $\mathbb{R}^d$ , equipped with the Borel  $\sigma$ -algebra.

A random variable is called *continuous* when its distribution does not have atoms. In particular, for a real-valued continuous random variable,  $\mathbb{P}(X = s) = 0$ ,  $\forall s \in \mathbb{R}$ .

The distribution of  $X$  is the push-forward of  $\mathbb{P}$  via  $X$ :

### 5.1.2 Absolutely Continuous and Singular Distributions

Let  $\mu_X = X_*(\mathbb{P})$  denote the distribution of  $X$ .

#### Definition 1. Absolutely Continuous

The distribution  $\mu_X$  is said to be *absolutely continuous* with respect to Lebesgue measure  $\lambda$  if there exists a non-negative function  $f \in L^1(\mathbb{R})$  such that  $\int_{\mathbb{R}} f d\lambda = 1$  and  $\mu_X(A) = \mathbb{P}(X \in A) = \int_A f(x) d\lambda(x)$  for every Borel set  $A \subseteq \mathbb{R}$ .

The function  $f$  is called the *density* of  $X$ .

#### Remark

The statement that the distribution of a random variable is absolutely continuous means that its distribution is absolutely continuous with respect to Lebesgue measure.

On the other hand, a distribution is called *singular* with respect to Lebesgue measure if it is concentrated on a set of Lebesgue measure zero.

More precisely, if  $\nu = X_*(\mathbb{P})$ , then  $\nu$  is singular with respect to Lebesgue measure if there exists a Borel set  $A \subseteq \mathbb{R}$  such that

$$\nu(A) = 1, \quad \lambda(A) = 0.$$

### 5.1.3 Examples of Continuous Random Variables

#### 1. Uniform random variable

The uniform distribution on  $[0, 1]$  has density

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

#### 2. Normal random variable

Let

$$X \sim N(\mu, \sigma^2), \quad \sigma > 0.$$

Its density is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Hence, for  $a < b$ ,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx.$$

#### 3. Exponential random variable

Let

$$\lambda > 0, \quad X \sim \text{Exp}(\lambda).$$

The random variable takes values in  $[0, \infty)$  and has density

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

Thus, for  $0 \leq a < b$ ,

$$\mathbb{P}(a < X < b) = \int_a^b \lambda e^{-\lambda x} dx.$$

The exponential distribution has the *memoryless property*:

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

### 5.1.4 A Singular Continuous Distribution

Consider  $\Omega = \{0, 1\}^{\mathbb{N}}$  with the product  $\sigma$ -algebra and the usual product probability measure.

Define

$$X(\omega) = \sum_{n=1}^{\infty} \frac{\omega_n}{2^n}.$$

Then

$$X : \{0, 1\}^{\mathbb{N}} \longrightarrow [0, 1].$$

#### Question 2.

What is the distribution of  $X$ ?

For a fixed sequence  $\tilde{\omega} : \{1, \dots, n\} \longrightarrow \{0, 1\}$ , consider the corresponding dyadic interval

$$I_{\tilde{\omega}} = \left[ \sum_{i=1}^n \frac{\tilde{\omega}_i}{2^i}, \sum_{i=1}^n \frac{\tilde{\omega}_i}{2^i} + \frac{1}{2^n} \right].$$

The event that the first  $n$  coordinates agree with  $\tilde{\omega}$  is the cylinder set  $[\tilde{\omega}]$ , and

$$\mathbb{P}(X \in I_{\tilde{\omega}}) = \mathbb{P}([\tilde{\omega}]) = 2^{-n}.$$

Since the length of the dyadic interval is also

$$|I_{\tilde{\omega}}| = 2^{-n},$$

we obtain, for dyadic intervals  $I \subseteq [0, 1]$ ,

$$\mathbb{P}(X \in I) = |I|.$$

This leads to the uniform distribution on  $[0, 1]$ .

### 5.1.5 The Cantor Distribution

Now consider

$$\tilde{X}(\omega) = \sum_{n=1}^{\infty} \frac{2\omega_n}{3^n}, \quad \omega_n \in \{0, 1\}.$$

This is the ternary expansion obtained by using only the digits 0 and 2.

Consequently,

$$\tilde{X}(\omega) \in C$$

almost surely, where  $C$  denotes the middle-thirds Cantor set.

Recall that

$$\lambda(C) = 0.$$

Thus the distribution of  $\tilde{X}$  is concentrated on a set of Lebesgue measure zero.

### Important

The Cantor distribution is a singular continuous distribution. It is continuous in the sense that

$$\mathbb{P}(\tilde{X} = \theta) = 0$$

for every  $\theta \in \mathbb{R}$ , but its distribution is singular with respect to Lebesgue measure because it is concentrated on the Cantor set.

To see why there are no atoms, consider the ternary expansion

$$\theta = \sum_{n=1}^{\infty} \frac{\theta_n}{3^n}.$$

If some  $\theta_i \notin \{0, 2\}$ , then

$$\mathbb{P}(\tilde{X} = \theta) = 0.$$

If  $\theta_i \in \{0, 2\}$  for every  $i$ , then fixing the first  $n$  digits determines a cylinder set of probability  $2^{-n}$ .

Therefore,

$$\mathbb{P}(\tilde{X} = \theta) \leq 2^{-n} \text{ for every } n, \text{ and hence } \mathbb{P}(\tilde{X} = \theta) = 0.$$

## 5.1.6 Bernoulli Convolution

Consider  $\lambda \in \left(\frac{1}{2}, 1\right)$  and let  $\omega = (\omega_1, \omega_2, \dots) \in \{-1, 1\}^{\mathbb{N}}$ .

Define

$$X_{\lambda}(\omega) = \sum_{n=1}^{\infty} \omega_n \lambda^n.$$

This is an example of a *Bernoulli convolution*.

### Question 3.

For which values of  $\lambda$  is the distribution of  $X_{\lambda}$  absolutely continuous?

This question is related to the classical problems surrounding Bernoulli convolutions and the Erdős conjecture.

**Question 4.**

What distribution should we use to model a given random phenomenon?

## 5.2 Bertrand's Paradox

**Question 5.**

Suppose we have a unit disc and an inscribed equilateral triangle. Consider a uniformly chosen chord of the disc.

What is the probability that the chord has length greater than the side length of the triangle?

Let  $AB$  denote the side of the inscribed equilateral triangle and  $CD$  denote the randomly chosen chord. We want to determine

$$\mathbb{P}(CD > AB).$$

The difficulty is that the phrase *uniformly chosen chord* does not specify a unique probability model.

**Remark**

Bertrand's paradox illustrates that the answer depends on how "uniformly" is interpreted.

### 5.2.1 Method 1: Choosing Two Endpoints

Choose the two endpoints  $C$  and  $D$  independently and uniformly from the circumference of the circle.

Under this model,

$$\mathbb{P}(CD > AB) = \frac{1}{3}.$$

### 5.2.2 Method 2: Choosing the Midpoint

Choose a radius uniformly and then choose a point  $M$  uniformly along that radius. Construct the unique chord whose midpoint is  $M$ .

Under this model,

$$\mathbb{P}(CD > AB) = \frac{1}{2}.$$

The difference in the two answers demonstrates the ambiguity in the phrase "uniformly chosen chord."

**Question 6.**

What is the distribution of the chord length under each of these probability models?

## 5.3 Distribution as a Push-Forward

The distribution of a random variable is simply the push-forward of the underlying probability measure.

Suppose

$$X : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

is a real-valued random variable.

Its distribution is

$$\nu = X_*(\mathbb{P}).$$

Therefore,

$$\mathbb{P}(X \in [a, b]) = \nu([a, b]) = \mathbb{P}\left(X^{-1}([a, b])\right).$$

### 5.3.1 Change of Variables

Suppose

$$\Omega = [a, b]$$

with Lebesgue probability measure, and suppose  $X$  is a sufficiently regular monotone function.

Then the distribution of  $X$  can be obtained from the push-forward measure.

If  $X$  is invertible, then

$$\mathbb{P}(X \in [c, d]) = \mathbb{P}\left(X^{-1}([c, d])\right).$$

For a monotone differentiable function,

$$X^{-1}(d) - X^{-1}(c) = \int_c^d (X^{-1})'(t) dt.$$

Using

$$(X^{-1})'(t) = \frac{1}{X'(X^{-1}(t))},$$

we obtain

$$\mathbb{P}(X \in [c, d]) = \int_c^d \frac{1}{X'(X^{-1}(t))} dt,$$

whenever the required regularity and monotonicity assumptions hold.

### 5.3.2 Example: Chord Length

#### Method 1.

Let

$$\Omega = [0, 1]$$

and define

$$X(\theta) = |e^{i\pi\theta} - 1|.$$

Since

$$|e^{i\alpha} - 1| = 2 \left| \sin \frac{\alpha}{2} \right|,$$

we have

$$X(\theta) = 2 \sin \left( \frac{\pi\theta}{2} \right), \quad 0 \leq \theta \leq 1.$$

Thus

$$X : [0, 1] \longrightarrow [0, 2].$$

Its inverse is

$$X^{-1}(x) = \frac{2}{\pi} \arcsin \left( \frac{x}{2} \right).$$

Therefore,

$$\frac{d}{dx} X^{-1}(x) = \frac{1}{\pi \sqrt{1 - \frac{x^2}{4}}}.$$

Hence the density of  $X$  is

$$f_X(x) = \frac{1}{\pi \sqrt{1 - \frac{x^2}{4}}}, \quad 0 < x < 2.$$

#### Method 2.

Alternatively, consider

$$X : [0, 1] \longrightarrow [0, 2]$$

defined by

$$X(\omega) = 2\sqrt{1 - \omega^2}.$$

Here,

$$X^{-1}(x) = \sqrt{1 - \frac{x^2}{4}}, \quad 0 \leq x \leq 2.$$

The corresponding density is obtained by differentiating the inverse appropriately.

**Important**

The central point of Bertrand's paradox is that specifying the word “uniformly” is not enough. A probability model must specify the underlying sample space and probability measure.

Different choices of the probability space and probability measure can produce different distributions for the same geometric quantity.

---

End of Lecture

---

# Probability Theory

## Lecture 06

### Introduction to Probability

**Date:** Friday, 21 August 2026

**Instructor:** Dr. Nishant Chandgotia

## 6.1 Distribution Functions and Random Variables

Random variables are measurable maps

$$X : (\Omega, \mathcal{B}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

We define the cumulative distribution function (C.D.F.) of  $X$  by

### Definition 1. Cumulative Distribution Function

The *cumulative distribution function* of a random variable  $X$  is the function

$$F_X : \mathbb{R} \longrightarrow [0, 1]$$

defined by

$$F_X(t) = \mathbb{P}(X \leq t).$$

### Remark

**Naming.** This is what Durrett calls the *distribution function*.

The C.D.F. completely determines the distribution of a real-valued random variable.

If  $X \sim \mu$ , then the C.D.F. of  $X$  is given by

$$F_X(t) = \mu((-\infty, t]).$$

Conversely, suppose that a C.D.F.  $F_X$  is given. We would like to construct the corresponding probability measure  $\mu_X$  on  $\mathbb{R}$ .

For  $a < b$ , define

$$\mu_X((a, b]) = F_X(b) - F_X(a).$$

One can check that this extends to a probability measure on  $\mathbb{R}$ .

### 6.1.1 Properties of the C.D.F.

Let  $F_X$  be the C.D.F. of a random variable  $X$ .

1.  $F_X$  is an increasing function.
2.  $\lim_{t \rightarrow \infty} F_X(t) = 1$ ,  $\lim_{t \rightarrow -\infty} F_X(t) = 0$ .
3.  $F_X$  is right-continuous:

$$\lim_{t \downarrow s} F_X(t) = F_X(s).$$

Indeed,

$$\lim_{t \downarrow s} F_X(t) = \lim_{t \downarrow s} \mathbb{P}(X \leq t) = \mathbb{P}(X \leq s),$$

by continuity from above.

#### Remark

The C.D.F. need not be left-continuous.

In fact,

$$\mathbb{P}(X = t) > 0$$

if and only if

$$F_X(t) - F_X(t^-) > 0,$$

where

$$F_X(t^-) = \lim_{s \uparrow t} F_X(s).$$

Moreover,

$$F_X(t) - F_X(t^-) = \mathbb{P}(X = t).$$

The remaining standard properties are:

4.  $F_X$  has at most countably many discontinuities.
5.  $F_X$  is Borel measurable.
6.  $F_X(t^-) = \mathbb{P}(X < t)$ .

## 6.2 The Probability Measure Associated with a C.D.F.

We now check the construction

$$\mu_X((a, b]) = F_X(b) - F_X(a)$$

and see why it defines a probability measure.

The natural collection of sets to consider first is the semiring

$$\mathcal{S} = \{(a, b] : a < b\}.$$

Suppose

$$(a, b] \subseteq \bigcup_{i=1}^{\infty} (a_i, b_i].$$

We want to verify the appropriate countable subadditivity/additivity property.

For each interval,

$$\mu_X((a_i, b_i]) = F_X(b_i) - F_X(a_i).$$

Hence

$$\sum_{i=1}^{\infty} \mu_X((a_i, b_i]) = \sum_{i=1}^{\infty} (F_X(b_i) - F_X(a_i)).$$

The key point is the right-continuity and monotonicity of  $F_X$ , which allow the interval  $(a, b]$  to be approximated by finitely many of the covering intervals.

### 6.2.1 Finite Additivity

Suppose

$$(a, b] = \bigcup_{i=1}^n (a_i, b_i]$$

where the intervals are disjoint and ordered so that

$$a = a_1 < b_1 = a_2 < b_2 = a_3 < \cdots < a_n < b_n = b.$$

Then

$$\begin{aligned} \mu_X((a, b]) &= F_X(b) - F_X(a) \\ &= \sum_{i=1}^n (F_X(b_i) - F_X(a_i)) \\ &= \sum_{i=1}^n \mu_X((a_i, b_i]). \end{aligned}$$

Thus  $\mu_X$  is finitely additive on the semiring  $\mathcal{S}$ .

### 6.2.2 Countable Additivity

The same idea extends to countable disjoint unions.

Suppose

$$(a, b] = \bigcup_{i=1}^{\infty} (a_i, b_i],$$

where the intervals are pairwise disjoint.

For the finite partial unions,

$$\bigcup_{i=1}^n (a_i, b_i] \subseteq (a, b],$$

and finite additivity gives

$$\sum_{i=1}^n \mu_X((a_i, b_i]) \leq \mu_X((a, b]).$$

Passing to the limit,

$$\sum_{i=1}^{\infty} \mu_X((a_i, b_i]) \leq \mu_X((a, b]).$$

The reverse inequality follows from the covering property, together with the continuity properties of  $F_X$ .

Consequently,

$$\mu_X((a, b]) = \sum_{i=1}^{\infty} \mu_X((a_i, b_i]).$$

Hence  $\mu_X$  is countably additive on the semiring.

## 6.3 Correspondence Between C.D.F.s and Probability Measures

Let

$$\mathcal{F} = \left\{ F : \begin{array}{l} F \text{ satisfies the properties} \\ \text{of a C.D.F.} \end{array} \right\}$$

and let

$$\mathcal{P} = \{\text{probability measures on } \mathbb{R}\}.$$

The correspondence  $F \mapsto \mu_F$  is bijective.

In other words, a probability measure on  $\mathbb{R}$  determines a unique C.D.F., and a C.D.F. determines a unique probability measure.

## 6.4 Constructing a Random Variable from a C.D.F.

### Question 1. Constructing a Random Variable from a C.D.F.

Suppose we are given a function  $F$  satisfying the properties of a C.D.F. Is there a random variable  $X$  such that

$$F = F_X?$$

We want to construct such a random variable explicitly.

The idea is to use the probability space

$$([0, 1], \mathcal{B}([0, 1]), \lambda),$$

where  $\lambda$  denotes Lebesgue measure.

### 6.4.1 Case 1: $F$ is Continuous and Strictly Increasing

Suppose first that

$$F : (\mathbb{R}) \longrightarrow (0, 1)$$

is bijective and has no discontinuities.

**Claim.** We can take

$$(\Omega, \mathcal{F}, \mathbb{P}) = ([0, 1], \mathcal{B}([0, 1]), \lambda).$$

For any probability measure  $\mu$ , we seek an algorithmic way of constructing

$$X : \Omega \longrightarrow \mathbb{R}$$

such that

$$X \sim \mu.$$

Define

$$X = F^{-1}.$$

Then, for every  $t \in \mathbb{R}$ ,

$$\begin{aligned} \mathbb{P}(X \leq t) &= \mathbb{P}(F^{-1} \leq t) \\ &= \mathbb{P}\left(\left\{\omega : F^{-1}(\omega) \leq t\right\}\right) \\ &= \mathbb{P}(\{\omega : \omega \leq F(t)\}) \\ &= \mathbb{P}((0, F(t)]) \\ &= F(t). \end{aligned}$$

Therefore,

$$F_X(t) = F(t),$$

and hence

$$F_X = F.$$

### 6.4.2 Case 2: $F$ Has a Jump

**Case 2.**

Suppose  $F$  has a jump at  $t$ .

Let

$$a = F(t^-), \quad b = F(t).$$

Define

$$X(\omega) = \begin{cases} F^{-1}(\omega), & \omega < a, \\ t, & a \leq \omega \leq b, \\ F^{-1}(\omega), & \omega > b. \end{cases}$$

Thus the entire interval  $(a, b]$  is mapped to the single point  $t$ .

This creates the required atom:

$$\mathbb{P}(X = t) = b - a = F(t) - F(t^-).$$

**Todo: Add the corresponding graph.**

### 6.4.3 Case 3: $F$ Has a Flat Portion

**Case 3.**

Suppose  $F$  is constant on an interval  $[a, b]$ .

Let

$$\theta = F(a) = F(b).$$

Define

$$X(\omega) = \begin{cases} F^{-1}(\omega), & \omega < \theta, \\ a, & \omega = \theta, \\ F^{-1}(\omega), & \omega > \theta. \end{cases}$$

The value of  $X$  at the single point  $\omega = \theta$  is irrelevant for the distribution, since

$$\lambda(\{\theta\}) = 0.$$

**Todo: Add the corresponding graph.**

### 6.4.4 Case 4: General C.D.F.

**Case 4.**

For a general C.D.F., define the generalized inverse by

**Definition 2. Generalized Inverse of a C.D.F.**

For a C.D.F.  $F$ , its generalized inverse is defined by

$$F^{-1}(\omega) = \inf \{x \in \mathbb{R} : F(x) \geq \omega\}, \quad \omega \in (0, 1).$$

Define

$$X(\omega) = F^{-1}(\omega).$$

We need to show that

$$\mathbb{P}(X \leq b) = F(b)$$

and that  $X$  is measurable.

Let

$$A_b = \{\omega : X(\omega) \leq b\}.$$

It is enough to show that

$$A_b = (0, F(b)].$$

**Claim.** For every  $b \in \mathbb{R}$ ,

$$\{\omega : F^{-1}(\omega) \leq b\} = (0, F(b)].$$

Consequently,

$$\begin{aligned}\mathbb{P}(X \leq b) &= \lambda(A_b) \\ &= \lambda((0, F(b)]) \\ &= F(b).\end{aligned}$$

Therefore,

$$F_X(b) = F(b)$$

for every  $b \in \mathbb{R}$ .

**Homework:** Prove the claim above and complete the remaining details of the construction.

---

End of Lecture

---

## 6.5 Quiz, 24 August 2026

Let  $\mathcal{R}$  denote the respective appropriate Borel sigma algebra, while  $m$  is the Lebesgue measure on  $(0, 1)$ .

### Quiz 3

1. Let  $F : \mathbb{R} \rightarrow (0, 1)$  be a function such that it is non-decreasing, right continuous, and

$$\lim_{x \rightarrow -\infty} F(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} F(x) = 1.$$

Prove that there is a probability measure  $\mu$  on  $(\mathbb{R}, \mathcal{R})$  such that

$$F(x) = \mu((-\infty, x]).$$

2. Let  $(\Omega, \mathcal{B}, \mu)$  be a probability space and  $X$  be a non-negative  $L^1$  random variable. If  $F_X$  is the c.d.f., prove that

$$\int_{\mathbb{R}} (1 - F_X(x)) dx = \mathbb{E}(X).$$

3. **This is for Bonus:** Let  $\mu$  be a probability measure on  $(\mathbb{R}^2, \mathcal{R})$ . Construct a random variable

$$X : ((0, 1), \mathcal{B}, m) \rightarrow \mathbb{R}^2$$

such that the distribution of  $X$  is  $\mu$ .

# Probability Theory

## Lecture 07

### Introduction to Probability

**Date:** Friday, 28 August 2026

**Instructor:** Dr. Nishant Chandgotia

## 7.1 Examples of Random Variables

### Example 1. Uniform Random Variable

Let  $X$  be uniformly distributed on  $(0, 1)$ . Then its C.D.F. is given by

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ x, & 0 < x < 1, \\ 1, & x \geq 1. \end{cases}$$

### Example 2. Exponential Random Variable

Let

$$X \sim \text{Exp}(\lambda), \quad \lambda > 0.$$

Then  $X$  has density

$$f_X(y) = \lambda e^{-\lambda y}, \quad y \geq 0.$$

### Example 3. Poisson Random Variable

Let

$$X \sim \text{Poisson}(\lambda), \quad \lambda > 0.$$

Then the C.D.F. of  $X$  is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \sum_{k=0}^{\lfloor x \rfloor} e^{-\lambda} \frac{\lambda^k}{k!}, & x \geq 0. \end{cases}$$

### Example 4. Standard Normal Random Variable

Let

$$X \sim N(0, 1).$$

Then the C.D.F. of  $X$  is

$$F_X(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy.$$

## 7.2 A Gaussian Tail Estimate

### Theorem 1. Gaussian Tail Bound

For  $x > 0$ ,

$$\left(\frac{1}{x} - \frac{1}{x^3}\right) e^{-x^2/2} \leq \int_x^\infty e^{-y^2/2} dy \leq \frac{1}{x} e^{-x^2/2}.$$

*Proof.* We first prove the upper bound.

Let

$$y = x + z.$$

Then

$$\int_x^\infty e^{-y^2/2} dy = e^{-x^2/2} \int_0^\infty e^{-xz - z^2/2} dz.$$

Since

$$e^{-xz - z^2/2} \leq e^{-xz},$$

we obtain

$$\begin{aligned} \int_x^\infty e^{-y^2/2} dy &\leq e^{-x^2/2} \int_0^\infty e^{-xz} dz \\ &= \frac{1}{x} e^{-x^2/2}. \end{aligned}$$

Thus,

$$\boxed{\int_x^\infty e^{-y^2/2} dy \leq \frac{1}{x} e^{-x^2/2}}.$$

□

Todo: We have to compute the lower bound also.

## 7.3 Expectation and Variance

### Definition 1. Expectation

Suppose  $X$  is a random variable such that

$$X \in L^1(\mathbb{P}).$$

Then the expectation of  $X$  is defined by

$$\mathbb{E}(X) = \int_{\Omega} X d\mathbb{P}.$$

**Definition 2. Variance**

Suppose

$$X \in L^2(\mathbb{P}).$$

Then the variance of  $X$  is defined by

$$\text{Var}(X) = \mathbb{E} \left[ (X - \mathbb{E}(X))^2 \right].$$

## 7.4 Central Limit Theorem

**Theorem 2. Central Limit Theorem**

Let  $X_1, X_2, \dots$  be i.i.d. random variables such that they all have same distribution say  $X$ ,  $\mathbb{E}(X_1^2) < \infty$ . Let

$$\mu = \mathbb{E}(X_1), \quad \sigma^2 = \text{Var}(X_1).$$

Then

$$\frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma} \xrightarrow{d} N(0, 1).$$

Equivalently, for  $a < b$ ,

$$\mathbb{P} \left( a \leq \frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma} \leq b \right) \rightarrow \frac{1}{\sqrt{2\pi}} \int_a^b e^{-x^2/2} dx.$$

**Remark**

The result above is the usual form of the Central Limit Theorem.

## 7.5 Examples of Expectation and Variance

**Example 5. Expectation and Variance of Common Distributions**

1. If  $X \sim \text{Exp}(\lambda)$ , then

$$\mathbb{E}(X) = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

2. If  $X \sim \text{Poisson}(\lambda)$ , then

$$\mathbb{E}(X) = \lambda, \quad \text{Var}(X) = \lambda.$$

3. If  $X \sim \text{Geom}(p)$ , then

$$\mathbb{E}(X) = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2}.$$

4. If  $X \sim N(\mu, \sigma^2)$ , then

$$\mathbb{E}(X) = \mu, \quad \text{Var}(X) = \sigma^2.$$

#### Question 1. Prove this

Do the calculations to prove  $\mathbb{E}(X) = \mu$  and  $\text{Var}(X) = \sigma^2$  for  $X \sim N(\mu, \sigma^2)$ .

#### Remark

If  $Y = aX + b$ , then  $\mathbb{E}(Y) = a\mathbb{E}(X) + b$  and  $\text{Var}(Y) = a^2 \text{Var}(X)$ .

## 7.6 A Result on Signs of Vectors

#### Theorem 3. Sign Choice for Vectors

Let  $v_1, \dots, v_n \in \mathbb{R}^n$  satisfy  $\|v_i\|_2 = 1$  for all  $i$ .

Then:

1. There exist  $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$  such that  $\|\sum_{i=1}^n \varepsilon_i v_i\|_2 \leq \sqrt{n}$ .
2. There exist  $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$  such that  $\|\sum_{i=1}^n \varepsilon_i v_i\|_2 \geq \sqrt{n}$ .

*Proof.* Let  $\varepsilon_1, \dots, \varepsilon_n$  be independent random variables, each uniformly distributed on  $\{-1, 1\}$ .

Then

$$\begin{aligned} \mathbb{E} \left[ \left\| \sum_{i=1}^n \varepsilon_i v_i \right\|_2^2 \right] &= \mathbb{E} \left[ \sum_{i,j} \varepsilon_i \varepsilon_j \langle v_i, v_j \rangle \right] \\ &= \sum_{i,j} \langle v_i, v_j \rangle \mathbb{E}(\varepsilon_i \varepsilon_j). \end{aligned}$$

For  $i \neq j$ ,  $\mathbb{E}(\varepsilon_i \varepsilon_j) = \mathbb{E}(\varepsilon_i) \mathbb{E}(\varepsilon_j) = 0$ , while  $\mathbb{E}(\varepsilon_i^2) = 1$ .

Therefore,

$$\begin{aligned} \mathbb{E} \left[ \left\| \sum_{i=1}^n \varepsilon_i v_i \right\|_2^2 \right] &= \sum_{i=1}^n \|v_i\|_2^2 \\ &= n. \end{aligned}$$

Hence there must exist a choice of signs for which  $\|\sum_{i=1}^n \varepsilon_i v_i\|_2^2 \geq n$ , and there must also exist a choice of signs for which  $\|\sum_{i=1}^n \varepsilon_i v_i\|_2^2 \leq n$ .

Taking square roots gives the result. □

## 7.7 Exercises on Probability Distributions

### Question 2

Suppose  $X$  is a discrete random variable such that

$$\mathbb{P}(X = k) = (1 - p)^{k-1}p, \quad k \geq 1, \quad k \in \mathbb{N}.$$

Find

$$\mathbb{P}(a < X < b).$$

### Question 3

Suppose  $X$  is a random variable whose probability density function is

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad \lambda > 0.$$

Find

$$\mathbb{P}(a < X < b).$$

## 7.8 Graph Theory

### Definition 3. Graph

A graph is a pair  $G = (V, E)$ , where

- $V$  is the set of vertices;
- $E$  is the set of edges.

In this lecture, we consider undirected graphs.

### Example 6. Basic Graphs

Examples of graphs include:

1. The complete simple graph  $K_n$ .
2. The cycle graph  $C_n$ , where  $V = \mathbb{Z}_n$  and  $E = \{(i, i + 1) : i \in \mathbb{Z}_n\}$ .

## 7.9 Bipartite Graphs

### Definition 4. Bipartite Graph

A graph  $G = (V, E)$  is called bipartite if there exists a partition  $V = V_1 \sqcup V_2$  such that whenever  $(u, v) \in E$ , we have  $u \in V_1, v \in V_2$ , or equivalently,  $u \in V_2, v \in V_1$ .

### Example 7. Odd Cycle is Not Bipartite

The cycle graph  $C_n$  is bipartite if and only if  $n$  is even.  
For an odd cycle, one cannot partition the vertices into two sets such that every edge joins vertices belonging to different sets.

### Example 8. Path Graph

The path graph  $P_n$  is always bipartite.

## 7.10 Subgraphs and Walks

### Definition 5. Subgraph

A graph  $G' = (V', E')$  is a subgraph of  $G = (V, E)$  if  $V' \subseteq V$  and  $E' \subseteq E$ .

### Definition 6. Walk

A walk from a vertex to another vertex in  $G = (V, E)$  is a sequence of vertices  $v_0, v_1, \dots, v_n$  such that  $(v_i, v_{i+1}) \in E$  for every  $i = 0, \dots, n-1$ .

### Definition 7. Tree

A graph is called a tree if it is connected and does not contain a cycle as a subgraph.

### Definition 8. Connected Graph

A graph is called connected if, for all distinct vertices  $u, v \in V$ , there exists a walk from  $u$  to  $v$ .

## 7.11 Trees and Bipartite Graphs

### Theorem 4. Every Tree is Bipartite

Every tree is a bipartite graph.

*Proof.* Let  $T = (V, E)$  be a tree and choose a vertex  $v \in V$ .

Define

$$V_0 = \{u \in V : d(u, v) \text{ is even}\}$$

and

$$V_1 = \{u \in V : d(u, v) \text{ is odd}\},$$

where  $d(u, v)$  denotes the length of the shortest walk from  $u$  to  $v$ .

Clearly,

$$V = V_0 \sqcup V_1.$$

Suppose  $(u, w) \in E$ . If  $d(u, v)$  and  $d(w, v)$  had the same parity, then combining shortest paths from  $v$  to  $u$  and  $v$  to  $w$  with the edge  $(u, w)$  would produce a cycle.

Since  $T$  is a tree, this is impossible.

Therefore every edge joins a vertex in  $V_0$  to a vertex in  $V_1$ . Hence  $T$  is bipartite.

□

## 7.12 Characterization of Bipartite Graphs

### Theorem 5. Characterization of Bipartite Graphs

A graph  $G = (V, E)$  is bipartite if and only if it contains no cycle of odd length.

#### Remark

An odd cycle is not bipartite.

Moreover, every subgraph of a bipartite graph is also bipartite.

*Proof.* We have already seen that every tree is bipartite.

Now suppose that  $G$  is bipartite. Let  $V = V_0 \sqcup V_1$  be a bipartition of  $V$ .

Consider a cycle

$$v_0, v_1, \dots, v_{n-1}, v_0.$$

Since every edge joins the two different parts of the bipartition, the vertices of the cycle must alternate between  $V_0$  and  $V_1$ .

Therefore the length of the cycle must be even.

Hence every cycle in a bipartite graph is even.

□

## 7.13 Bipartite Subgraph

### Theorem 6. Bipartite Subgraph

Let  $G = (V, E)$  be a finite graph without self-loops. Then there exists a bipartite subgraph  $G' = (V, E')$  such that  $|E'| \geq \frac{|E|}{2}$ .

*Proof.* Let  $V$  be the vertex set and consider a random partition  $V = V_0 \sqcup V_1$  obtained by independently assigning each vertex to  $V_0$  or  $V_1$  with probability  $1/2$ .

An edge  $(u, v) \in E$  is said to be *crossing* if  $u$  and  $v$  belong to different parts of the partition.

Define the random variable

$$X_{(u,v)} = \begin{cases} 1, & \text{if } (u, v) \text{ is crossing,} \\ 0, & \text{otherwise.} \end{cases}$$

Let  $X = \sum_{(u,v) \in E} X_{(u,v)}$  be the total number of crossing edges.

For a fixed edge  $(u, v)$ ,

$$\begin{aligned} \mathbb{E}(X_{(u,v)}) &= \mathbb{P}(X_{(u,v)} = 1) \\ &= \mathbb{P}(u \in V_0, v \in V_1) + \mathbb{P}(u \in V_1, v \in V_0) \\ &= \frac{1}{4} + \frac{1}{4} \\ &= \frac{1}{2}. \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbb{E}(X) &= \sum_{(u,v) \in E} \mathbb{E}(X_{(u,v)}) \\ &= \frac{|E|}{2}. \end{aligned}$$

Hence there must exist at least one partition for which

$$X \geq \frac{|E|}{2}.$$

Take  $E'$  to be the set of crossing edges for such a partition. Then  $G' = (V, E')$  is bipartite and satisfies  $|E'| \geq \frac{|E|}{2}$ .

□

---

End of Lecture

---

# Probability Theory

## Lecture 08

### Introduction to Probability

**Date:** Wednesday, 2 September 2026

**Instructor:** Dr. Nishant Chandgotia

Suppose  $X, Y \in L^1(\mathbb{P})$  and  $X$  and  $Y$  are independent. Then  $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ .

**Question 1. Why does this make sense?**

Why does  $XY \in L^1(\mathbb{P})$ ?

#### Remark

The basic idea is that if  $X \sim \mu$  and  $Y \sim \nu$  are independent, then

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B) = \mu(A)\nu(B).$$

Hence,  $(X, Y) \sim \mu \times \nu$ .

Therefore,

$$\begin{aligned}\mathbb{E}(|XY|) &= \mathbb{E}(|X||Y|) \\ &= \int_{\mathbb{R}^2} |x||y| d(\mu \times \nu)(x, y).\end{aligned}$$

By Fubini's theorem,

$$\begin{aligned}\mathbb{E}(|XY|) &= \left( \int_{\mathbb{R}} |x| d\mu(x) \right) \left( \int_{\mathbb{R}} |y| d\nu(y) \right) \\ &= \mathbb{E}(|X|)\mathbb{E}(|Y|) < \infty.\end{aligned}$$

The same argument applied to  $XY$  gives

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y).$$

This also works when  $X, Y \geq 0$ .

Similarly, if  $X_1, X_2, \dots, X_n \in L^1(\mathbb{P})$  are independent, then

$$\mathbb{E}\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n \mathbb{E}(X_i).$$

#### Definition 1. Covariance

For random variables  $X, Y \in L^2(\mathbb{P})$ , the covariance of  $X$  and  $Y$  is defined by

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))].$$

Equivalently,

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

**Remark**

If  $X$  and  $Y$  are independent, then  $\text{Cov}(X, Y) = 0$ .  
The converse is not true in general.

**Example 1. Uncorrelated but Dependent Random Variables**

Consider random variables  $X, Y$  taking values in  $\{-1, 0, 1\}$  with joint probability table

|          | $Y = 1$ | $Y = 0$ | $Y = -1$ |
|----------|---------|---------|----------|
| $X = 1$  | 0       | $a$     | 0        |
| $X = 0$  | $b$     | $c$     | $b$      |
| $X = -1$ | 0       | $a$     | 0        |

where  $2a + 2b + c = 1$ ,  $a, b, c \geq 0$ .

Clearly this is a probability distribution.

We have  $\mathbb{P}(X = 1, Y = 1) = 0$ , but  $\mathbb{P}(X = 1)\mathbb{P}(Y = 1) > 0$  whenever  $a, b > 0$ . Hence  $X$  and  $Y$  are not independent.

On the other hand,  $\mathbb{E}(X) = 0$ ,  $\mathbb{E}(Y) = 0$ ,  
and therefore

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = 0.$$

Thus zero covariance does not imply independence.

**Question 2. Distribution of a Transformation**

If  $X \sim \text{Unif}[-1, 1]$ , what is the distribution of  $X^3$ ?

**Question 3. Bernoulli Random Variables**

Suppose  $X, Y \sim \text{Ber}$ .

What conditions on  $X$  and  $Y$  imply independence?

**Question 4. Jointly Normal Random Variables**

Suppose  $X$  and  $Y$  are jointly normal random variables.

Does  $\text{Cov}(X, Y) = 0$  imply that  $X$  and  $Y$  are independent?

**Question 5. Bernoulli Variables and Covariance**

Suppose  $X \sim \text{Ber}(p)$ ,  $Y \sim \text{Ber}(q)$ .

If  $\text{Cov}(X, Y) = 0$ , does it follow that  $X$  and  $Y$  are independent?

## 8.1 Inequalities

### Definition 2. Markov's Inequality

Let  $X \geq 0$  be a random variable and let  $a > 0$ . Then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}(X)}{a}.$$

### Remark

The main idea behind Markov's inequality is the following. Let  $\varphi : \mathbb{R} \rightarrow \mathbb{R}$  be a measurable function such that  $\varphi \geq 0$ .

For  $A \in \mathcal{F}$ ,

$$\mathbb{E}[\varphi(X)\mathbf{1}_{\{X \in A\}}] \leq \mathbb{E}[\varphi(X)].$$

This simple inequality has many important consequences.

### Definition 3. Chebyshev's Inequality

Let  $\mu = \mathbb{E}(X)$ ,  $\sigma^2 = \text{Var}(X)$ ,  $\lambda > 0$ .

Then

$$\mathbb{P}(|X - \mu| \geq \lambda\sigma) \leq \frac{1}{\lambda^2}.$$

### Remark

Let,  $Y = (X - \mu)^2$ . Then  $\mathbb{E}(Y) = \text{Var}(X) = \sigma^2$ .

Therefore, by Markov's inequality,

$$\begin{aligned} \mathbb{P}(|X - \mu| \geq \lambda\sigma) &= \mathbb{P}(Y \geq \lambda^2\sigma^2) \\ &\leq \frac{\sigma^2}{\lambda^2\sigma^2} \\ &= \frac{1}{\lambda^2}. \end{aligned}$$

### Definition 4. Chernoff Bound

For  $t > 0$ , define the moment generating function

$$M(t) = \mathbb{E}(e^{tX}).$$

Then

$$\mathbb{P}(X \geq a) \leq \frac{M(t)}{e^{ta}}.$$

**Remark**

The Chernoff bound follows immediately by applying Markov's inequality to the nonnegative random variable  $e^{tX}$ :

$$\begin{aligned}\mathbb{P}(X \geq a) &= \mathbb{P}(e^{tX} \geq e^{ta}) \\ &\leq \frac{\mathbb{E}(e^{tX})}{e^{ta}} \\ &= \frac{M(t)}{e^{ta}}.\end{aligned}$$

## 8.2 Coupon Collector Problem

**Question 6. Coupon Collector Problem**

Suppose there are  $n$  different types of cards. Each time a card is picked, each of the  $n$  types is obtained independently and uniformly.

How many cards do we need to draw to ensure that we have obtained all  $n$  types of cards?

Consider the probability space  $\Omega = (\{1, 2, \dots, n\}^{\mathbb{N}}, \mathcal{B}, \text{Unif}^{\otimes \mathbb{N}})$ .

Let  $X_n(\omega) = \#\{\text{cards picked to obtain the full collection}\}$ .

Equivalently,

$$X_n(\omega) = \min \{m : \{\omega_1, \omega_2, \dots, \omega_m\} = \{1, \dots, n\}\}.$$

**Theorem 1. Coupon Collector Bound**

For every  $\varepsilon > 0$ ,

$$\mathbb{P}\left(1 - \varepsilon < \frac{X_n}{n \log n} < 1 + \varepsilon\right) \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

*Proof.* We prove the upper bound. It is enough to show that

$$\mathbb{P}\left(\frac{X_n}{n \log n} > 1 + \varepsilon\right) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Equivalently,

$$\mathbb{P}(X_n > (1 + \varepsilon)n \log n) \rightarrow 0.$$

Let  $Y_k$  denote the number of card types missing after  $k$  draws. Thus,

$$Y_k(\omega) = n - |\{\omega_1, \omega_2, \dots, \omega_k\}|.$$

We want to show that if  $k > (1 + \varepsilon)n \log n$ , then  $Y_k$  is very likely to be zero.

For each  $i \in \{1, \dots, n\}$  and each  $k \geq 1$ , define

$$S_{i,k} = \begin{cases} 1, & \text{if card type } i \text{ is not picked by time } k, \\ 0, & \text{otherwise.} \end{cases}$$

Then  $Y_k = \sum_{i=1}^n S_{i,k}$ .

For a fixed  $i$ ,  $\mathbb{E}(S_{i,k}) = \left(1 - \frac{1}{n}\right)^k$ , since the probability that type  $i$  is not selected in one draw is  $1 - \frac{1}{n}$ .

Therefore,

$$\mathbb{E}(Y_k) = n \left(1 - \frac{1}{n}\right)^k.$$

Now take  $k = (1 + \varepsilon)n \log n$ .

By Markov's inequality,

$$\begin{aligned} \mathbb{P}(X_n > (1 + \varepsilon)n \log n) &= \mathbb{P}(Y_{(1+\varepsilon)n \log n} \geq 1) \\ &\leq \mathbb{E}[Y_{(1+\varepsilon)n \log n}] \\ &= n \left(1 - \frac{1}{n}\right)^{(1+\varepsilon)n \log n}. \end{aligned}$$

Using  $1 - \frac{1}{n} \leq e^{-1/n}$ , we obtain

$$\begin{aligned} n \left(1 - \frac{1}{n}\right)^{(1+\varepsilon)n \log n} &\leq n \exp\left(-\frac{(1 + \varepsilon)n \log n}{n}\right) \\ &= n e^{-(1+\varepsilon) \log n} \\ &= n^{-\varepsilon}. \end{aligned}$$

Hence,

$$\mathbb{P}(X_n > (1 + \varepsilon)n \log n) \leq n^{-\varepsilon} \longrightarrow 0.$$

Thus,

$$\mathbb{P}\left(\frac{X_n}{n \log n} > 1 + \varepsilon\right) \longrightarrow 0.$$

This proves the desired upper bound. □

## 8.3 Quiz, 4 September 2026

### Quiz 4

1. It is equivalent to a question discussed in class. What is it?
2. If  $X \sim \text{Geo}(p)$ , prove that  $\mathbb{E}(X) = \frac{1}{p}$ ,  $\text{Var}(X) \leq \frac{1}{p^2}$ .
3. Let  $X_k^{(n)}$  be the time taken for  $k$  bins to be filled. Prove that  $S_k^{(n)} = X_{k+1}^{(n)} - X_k^{(n)} \sim \text{Geo}(p_k)$ , where  $p_k = 1 - \frac{k}{n}$ , and that the random variables  $S_0^{(n)}, S_1^{(n)}, \dots, S_{n-1}^{(n)}$  are independent.
4. Prove that  $\mathbb{E}(X_n^{(n)}) = n \sum_{m=1}^n \frac{1}{m}$ .
5. Prove that  $\text{Var}(X_n^{(n)}) \leq cn^2$  for some constant  $c$ .
6. Prove that  $\frac{\mathbb{E}(X_n^{(n)})}{n \log n} \rightarrow 1$  as  $n \rightarrow \infty$ .
7. Use Chebyshev's inequality to conclude that  $\frac{X_n^{(n)} - n \sum_{m=1}^n \frac{1}{m}}{nf(n)} \rightarrow 0$  in probability as  $f(n) \rightarrow \infty$ .

# Probability Theory

## Lecture 09

### Introduction to Probability

**Date:** Monday, 7 September 2026

**Instructor:** Dr. Nishant Chandgotia

## 9.1 Continuation of Coupon Collector Problem

We have already defined  $X_n$  and  $Y_k = \sum_{i=1}^n S_{i,k}$ .

Now,

$$\text{Var}(Y_k) = \sum_{i=1}^n \text{Var}(S_{i,k}) + \sum_{\substack{i,j=1 \\ i \neq j}}^n \text{Cov}(S_{i,k}, S_{j,k}).$$

We have

$$\text{Var}(S_{i,k}) = \left(1 - \frac{1}{n}\right)^k \left[1 - \left(1 - \frac{1}{n}\right)^k\right].$$

Now,

$$\begin{aligned} \text{Cov}(S_{i,k}, S_{j,k}) &= \mathbb{E}(S_{i,k} S_{j,k}) - \mathbb{E}(S_{i,k}) \mathbb{E}(S_{j,k}) \\ &= \left(1 - \frac{2}{n}\right)^k - \left(1 - \frac{1}{n}\right)^{2k}. \end{aligned}$$

Since

$$(1 - x)^2 \geq 1 - 2x,$$

we have

$$\left(1 - \frac{1}{n}\right)^2 \geq 1 - \frac{2}{n},$$

and hence

$$\left(1 - \frac{2}{n}\right)^k \leq \left(1 - \frac{1}{n}\right)^{2k}.$$

Therefore,

$$\text{Cov}(S_{i,k}, S_{j,k}) \leq 0.$$

Thus,

$$\text{Var}(Y_k) \leq n \left(1 - \frac{1}{n}\right)^k \left[1 - \left(1 - \frac{1}{n}\right)^k\right].$$

Also,

$$\mathbb{E}(Y_k) = n \left(1 - \frac{1}{n}\right)^k.$$

Therefore,

$$\begin{aligned} \frac{\mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}} &\geq \frac{n \left(1 - \frac{1}{n}\right)^k}{\sqrt{n \left(1 - \frac{1}{n}\right)^k \left[1 - \left(1 - \frac{1}{n}\right)^k\right]}} \\ &\geq \sqrt{n \left(1 - \frac{1}{n}\right)^k}. \end{aligned}$$

### Question 1. Coupon Collector Problem

We want to show that

$$\mathbb{P}(X_n < (1 - \epsilon)n \log n) \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Consider

$$k = (1 - \epsilon)n \log n.$$

Then

$$\mathbb{P}(X_n < k) = \mathbb{P}(Y_k = 0).$$

By Chebyshev's inequality,

$$\begin{aligned} \mathbb{P}(Y_k = 0) &\leq \mathbb{P}\left(\left|\frac{Y_k - \mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}}\right| \geq \frac{\mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}}\right) \\ &\leq \frac{\text{Var}(Y_k)}{\mathbb{E}(Y_k)^2}. \end{aligned}$$

More generally, Chebyshev gives

$$\mathbb{P}\left(\left|\frac{Y_k - \mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}}\right| \geq t\right) \leq \frac{1}{t^2}.$$

We have

$$|Y_k - \mathbb{E}(Y_k)| \geq t\sqrt{\text{Var}(Y_k)} \iff Y_k \notin \left(\mathbb{E}(Y_k) - t\sqrt{\text{Var}(Y_k)}, \mathbb{E}(Y_k) + t\sqrt{\text{Var}(Y_k)}\right).$$

Choose  $t$  such that

$$0 < \mathbb{E}(Y_k) - t\sqrt{\text{Var}(Y_k)}.$$

Thus,

$$t \leq \frac{\mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}}.$$

If

$$Y_k = 0,$$

then

$$\begin{aligned} \left|\frac{Y_k - \mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}}\right| &= \frac{\mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}} \\ &\geq \sqrt{n \left(1 - \frac{1}{n}\right)^k}. \end{aligned}$$

For

$$k = (1 - \epsilon)n \log n,$$

we have

$$\begin{aligned} n \left(1 - \frac{1}{n}\right)^k &\sim n e^{-(1-\epsilon) \log n} \\ &= n^\epsilon. \end{aligned}$$

Hence,

$$\frac{\mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}} \geq \sqrt{n^\epsilon} = n^{\epsilon/2}.$$

Therefore,

$$\mathbb{P}(Y_{(1-\epsilon)n \log n} = 0) \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

**Todo:** Check and fill the corresponding graphs.

## 9.2 Erdős–Rényi Graphs

Let  $n \in \mathbb{N}$ ,  $p \in (0, 1)$ . The random graph  $G(n, p)$  is a random graph with  $n$  vertices  $\{1, 2, \dots, n\}$ , where each edge is chosen independently with probability  $p$ .

### Question 2. Number of Copies of $K_r$

How many copies of  $K_r$  are there in  $G(n, p)$ , where  $K_r$  denotes the complete graph on  $r$  vertices?

Let  $A \subseteq \{1, 2, \dots, n\}$  have cardinality  $r$ .

Define  $X_A = \begin{cases} 1, & \text{if } K_r \text{ appears on } A, \\ 0, & \text{otherwise.} \end{cases}$

Let  $X$  denote the number of copies of  $K_r$ . Then

$$X = \sum_{\substack{A \subseteq \{1, 2, \dots, n\} \\ |A|=r}} X_A.$$

Since all the edges of  $K_r$  must be present,

$$\mathbb{E}(X_A) = p^{\binom{r}{2}}.$$

Therefore,

$$\begin{aligned} \mathbb{E}(X) &= \sum_{\substack{A \subseteq \{1, 2, \dots, n\} \\ |A|=r}} \mathbb{E}(X_A) \\ &= \binom{n}{r} p^{\binom{r}{2}}. \end{aligned}$$

Now,

$$\begin{aligned}\text{Var}(X) &= \sum_{\substack{A \subseteq \{1,2,\dots,n\} \\ |A|=r}} \text{Var}(X_A) \\ &\quad + \sum_{\substack{A \neq B \\ |A|=|B|=r}} \text{Cov}(X_A, X_B).\end{aligned}$$

For the first term,

$$\sum_{|A|=r} \text{Var}(X_A) = \binom{n}{r} p^{\binom{r}{2}} (1 - p^{\binom{r}{2}}).$$

If  $|A \cap B| \leq 1$ , then the sets of edges involved in  $X_A$  and  $X_B$  are disjoint.

Hence,

$$\text{Cov}(X_A, X_B) = 0.$$

If  $|A \cap B| = t > 1$ , then

$$\begin{aligned}\text{Cov}(X_A, X_B) &= \mathbb{E}(X_A X_B) - \mathbb{E}(X_A) \mathbb{E}(X_B) \\ &= p^{2\binom{r}{2} - \binom{t}{2}} - p^{2\binom{r}{2}}.\end{aligned}$$

Thus,

$$\sum_{\substack{A \neq B, |A \cap B|=t \\ |A|=|B|=r}} \text{Cov}(X_A, X_B) = \binom{n}{r} \binom{n-r}{r-t} \left[ p^{2\binom{r}{2} - \binom{t}{2}} - p^{2\binom{r}{2}} \right].$$

Consequently,

$$\begin{aligned}\text{Var}(X) &= \binom{n}{r} p^{\binom{r}{2}} (1 - p^{\binom{r}{2}}) \\ &\quad + \sum_{t=2}^{r-1} \binom{n}{r} \binom{n-r}{r-t} \left[ p^{2\binom{r}{2} - \binom{t}{2}} - p^{2\binom{r}{2}} \right].\end{aligned}$$

For some constants  $c_1, c_2 > 0$ ,

$$\mathbb{E}(X) \sim c_1 n^r, \quad \text{Var}(X) \sim c_2 n^{2r-2}.$$

By Chebyshev's inequality, we have

$$\mathbb{P} \left( \left| \frac{X - \mathbb{E}(X)}{n^{r-1} f(n)} \right| > \epsilon \right) \longrightarrow 0 \quad \text{as } n \rightarrow \infty,$$

where  $f(n)$  is any function such that  $f(n) \longrightarrow \infty$  as  $n \rightarrow \infty$ .

In fact,

$$\begin{aligned}\mathbb{P}(|X - \mathbb{E}(X)| > \epsilon \mathbb{E}(X)) &\leq \frac{\text{Var}(X)}{\epsilon^2 \mathbb{E}(X)^2} \\ &\sim \frac{1}{n^2 \epsilon^2} \longrightarrow 0.\end{aligned}$$

### 9.2.1 Full Strength of Chebyshev

Let  $X_1, X_2, \dots, X_n$  be random variables and define  $S_n = \sum_{i=1}^n X_i$ .

Suppose  $\mathbb{E}(X_i) = \mu$ ,  $\text{Var}(X_i) \leq C$  for all  $i$ , and

$$\text{Cov}(X_i, X_j) = 0 \quad \text{for all } i < j.$$

Thus, the random variables are uncorrelated.

Then  $\frac{S_n}{n} \rightarrow \mu$  in  $L^2$  and in probability.

This is a version of the weak law of large numbers.

**Proof.**

We have

$$\begin{aligned} \mathbb{E} \left[ \left| \frac{S_n}{n} - \mu \right|^2 \right] &= \text{Var} \left( \frac{S_n}{n} \right) \\ &= \frac{1}{n^2} \text{Var}(S_n). \end{aligned}$$

Since the random variables are uncorrelated,

$$\begin{aligned} \text{Var}(S_n) &= \sum_{i=1}^n \text{Var}(X_i) \\ &\leq Cn. \end{aligned}$$

Therefore,

$$\mathbb{E} \left[ \left| \frac{S_n}{n} - \mu \right|^2 \right] \leq \frac{Cn}{n^2} = \frac{C}{n} \rightarrow 0.$$

Hence,  $\frac{S_n}{n} \rightarrow \mu$  in  $L^2$ , and therefore  $\frac{S_n}{n} \rightarrow \mu$  in probability.

**But further:**

By Chebyshev's inequality,

$$\begin{aligned} \mathbb{P} \left( \left| \frac{S_n}{n} - \mu \right| > \epsilon \mu \right) &\leq \frac{\text{Var}(S_n)}{(n\epsilon\mu)^2} \\ &\leq \frac{nC}{n^2\epsilon^2\mu^2} \\ &= \frac{C}{n\epsilon^2\mu^2} \rightarrow 0. \end{aligned}$$

#### Question 3. Exponential Decay

How do we get exponential decay?

---

End of Lecture

---

# Feedback and Corrections

These notes are prepared as lecture notes and may contain typographical errors, omissions, or mathematical mistakes. If you notice any such errors or have suggestions for improving the notes, please feel free to let me know.

Your feedback is greatly appreciated and will help improve future versions of these notes.

[Report an error or suggest a correction](#)

# References

- [Cha26] Nishant Chandgotia. *Personal Website*. <https://nishantchandgotia.github.io/>. Accessed: 2026-08-05. 2026.
- [Cha] Nishant Chandgotia. *Probability Rough Lecture Notes*. Updated Regularly. URL: <https://drive.google.com/file/d/1lItA2nWPtJzKppXzdLaK5lwnQJC5VrzH/view>.
- [Dur19] Rick Durrett. *Probability: Theory and Examples*. 5th ed. Cambridge University Press, 2019.
- [Fel68] William Feller. *An Introduction to Probability Theory and Its Applications, Volume I*. 3rd ed. Wiley, 1968.
- [Fel71] William Feller. *An Introduction to Probability Theory and Its Applications, Volume II*. 2nd ed. Wiley, 1971.
- [Gho26] Debjit Ghoshal. *Handout 1*. Shared on August 3. 2026. URL: [https://drive.google.com/file/d/170rabd4lwC6C\\_UWW3V52bXt3YIzxx8Cg/view](https://drive.google.com/file/d/170rabd4lwC6C_UWW3V52bXt3YIzxx8Cg/view).
- [Hai03] John H. Haigh. *Taking Chances: Winning with Probability*. 2nd ed. Also published under the title “Knowing the Odds” in some editions/regions. Oxford University Press, 2003.
- [Kri] Manjunath Krishnapur. *Probability Notes*. Lecture Notes. URL: <https://math.iisc.ac.in/~manju/PT2019/probabilitytheory.html>.
- [Wika] Wikimedia Commons. *Galton Board*. <https://upload.wikimedia.org/wikipedia/commons/d/d2/GaltonBoard.png>. Accessed August 12, 2026.
- [Wikb] Wikipedia. *Monty Hall Problem*. Online Resource. URL: [https://en.wikipedia.org/wiki/Monty\\_Hall\\_problem](https://en.wikipedia.org/wiki/Monty_Hall_problem).
- [Wikc] Wikipedia. *Sleeping Beauty Problem*. Online Resource. URL: [https://en.wikipedia.org/wiki/Sleeping\\_Beauty\\_problem](https://en.wikipedia.org/wiki/Sleeping_Beauty_problem).
- [Wil91] David Williams. *Probability with Martingales*. Cambridge University Press, 1991.